Magnetotelluric static shift correction using an equivalent source technique

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SUMMARY

The static shift caused by near-surface conductors poses a significant challenge in magnetotelluric (MT) inversion. We present a technique to remove the static shifts in MT data by using the equivalent source technique. This method is based on the assumption that all the data are measured on or above the surface. Since static shifts across multiple stations at the same frequency are in general uncorrelated, they can be identified and removed by constructing an equivalent source layer that reproduces only the coherent signal in the data observed at multiple stations. Here we use a synthetic three-dimensional model to demonstrate the technique and demonstrate the effectiveness of the method using field data sets in the presentation.

INTRODUCTION

In magnetotelluric surveys local inhomogeneities often distort the measured impedance responses. The distortions can be both inductive and galvanic (Berdichevsky and Dimitriev, 1976). When the dimensions of superficial structures are much smaller than the skin depth, inductive distortion is negligible below a certain frequency. The galvanic distortion is caused by accumulated charges on the boundaries of conductors or across conductivity gradients. It persists at all frequencies. Therefore, the galvanic effect is the most significant problem when using MT for deep structure soundings.

The galvanic effect primarily affects the electric field and we therefore focus upon it. The distortion usually causes apparent resistivity curves to shift on the standard log-apparent-resistivity versus log-frequency display, whereas the phase curves are less affected. This phenomenon is termed the static shift.

In the past three decades, a large amount of work has been devoted to address static shift in MT. The methods proposed include curve shifting, spatial filtering, distortion tensor decomposition, and inversion. Curve shifting is based on the assumption that the apparent resistivity curves should maintain the shape. A reference regional curve is found and all the other curves are shifted toward the reference curve by their differences. For example, Sternberg et al. (1988) used transient electromagnetic (TEM) sounding curves of the shallow subsurface to produce reference curves and shifted the MT apparent resistivity curves. Spatial filtering makes use of a low-pass spatial filter to remove static shifts. One of the most widely-used spatial filtering methods is electromagnetic array profiling (EMAP), first proposed by Bostick (1986). Groom and Bailey (1989) presented a distortion tensor decomposition method that can separate the tensor into determinable and indeterminable parts. deGroot Hedlin (1991) and Ogawa and Uchida (1996) both performed a straightforward inversion by including static shifts as inversion parameters based on statistical assumptions about static shifts. However, the effectiveness of each method is dependent on the geo-electrical environment. Static shift remains a challenge in MT interpretation. More comprehensive reviews can be found in Jiracek (1990) and Bibby et al. (2005).

We propose an alternative method to correct MT static shift by using the equivalent source technique (Dempney, 1969), which was first developed in gravity and magnetic data processing. MacLennan and Li (2013) used this approach to remove static shift in controlled source electromagnetic (CSEM) data. The equivalent source technique works with fields, so a crucial step is to obtain the electrical component in MT for processing by this technique, because MT impedance data are the ratios of electric fields over magnetic fields. We accomplish this by deriving a 1-D regional structure and extracting the electrical field components from the impedance data through the magnetic fields associated with two different polarization modes. Denoised impedance data are then reconstructed from the denoised electrical fields.

METHOD

In MT, we typically measure two horizontal electric field components and three magnetic field components on the surface, and derive the impedance data as the ratios of electric and magnetic fields. The electric field satisfies Helmholtz equation,

$$\bigtriangledown^2\vec{E}+(\mu\varepsilon\omega^2+i\omega\mu\sigma)\vec{E}=0, \tag{1}$$

where we assume a $e^{-i\omega t}$ time dependence. Under the quasi-static approximation, we neglect the second-order term in frequency (displacement current). Furthermore, the electric field satisfies the Laplace's equation in the air ($\sigma=0$),

$$\bigtriangledown^2\vec{E}=0. \tag{2}$$

By continuity, the horizontal components of the electric field measured on the surface is a harmonic function. Therefore, we can use the equivalent source technique to process the electric field data.

Equivalent source

The aim of equivalent source, introduced by Dempney (1969), is to construct a fictitious layer below the observation surface that can replicate the observed field within the source-free region, which is the free-air region in our case. If the values of the field on the boundary are given, then the field elsewhere within the source-free region is determined uniquely by solving a Dirichlet problem. Thus, if we can construct the equiva-

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lent source that reproduces the observed field on the boundary, then it will describe the field in the source-free region.

Our goal here is to reproduce only the coherent signal from the desired target and exclude the static shift caused by near-surface inhomogeneities. To achieve this, we formulate the equivalent construction as a regularized inverse problem. By finding the optimal regularization parameter, we seek to estimate the static shift level and thus remove it.

For our application, we choose a layer of continuous rectangular prisms as the equivalent source. Each cell has a constant electrical polarization P. For simplicity, we orient electrical polarization in the vertical direction. From our numerical experiments, the horizontal size of the cells should be comparable to the station spacing. The depth of the top of the layer should be much greater than the receiving station spacing. We also extend the equivalent source area horizontally to a minimum of three times the total extent of the data area. The thickness of the layer does not seem to affect the result significantly if it is placed sufficiently deep, and we set it to be 2 times the receiving station spacing for the following illustration.

Equivalent source construction using regularized inversion

We start by considering the horizontal components of the electric field. Here we define $\hat{X}$ toward north, $\hat{Y}$ toward east and $\hat{Z}$ vertical downward. The relationship between observed data and the equivalent source values is linear:

$$\vec{d}^{obs} = \mathbf{G}\vec{m}, \tag{3}$$

where $\vec{d}^{obs} = [d_x^1, ...d_x^q, d_y^1, ...d_y^q]^T$ are the observed electric field, $\mathbf{G}$ the sensitivity matrix, $\vec{m}$ the model representing the values of the equivalent source, q the number of data points, $N = 2q$ is the length of the data vector, M is the size of the discretized equivalent source. We note that $\vec{d}^{obs}$ and $\vec{m}$ are complex-valued, but $\mathbf{G}$ is real.

The electric field at any location $\vec{r}_i$ due to vertical electrical polarization P in a volume V is given by,

$$E_k^i = \sum_{j=1}^{M} P_j G_{ij}^k. \tag{4}$$

We assume that P is uniform in each cell of the equivalent source. The elements of the sensitivity matrix $\mathbf{G}$ are given by,

$$G_{ij}^k = \frac{1}{4\pi\varepsilon_0}\int_{V_j}\frac{\partial}{\partial x_k}\frac{\partial}{\partial x_3}\frac{1}{|\vec{r}_i - \vec{r}\,'|}dv, \tag{5}$$

where ε_0 is the permittivity of free space, and $k = 1, 2, 3$ indicate the three axis directions. The expression for the above integral over a rectangular prism cell is given by Sharma (1966).

The observed data contains coherent signal from the desired target and static shift. The static shifts at each station across frequencies are correlated, whereas the static shifts across multiple stations at any given frequency are uncorrelated. Therefore, we process the data frequency by frequency to remove the uncorrelated static shifts.

We construct the equivalent source by formulating the process as a linear inverse problem using Tikhonov regularization. We minimize the following objective function,

$$\phi = \phi_d + \beta\phi_m, \tag{6}$$

where ϕ_d is the data misfit function defined by,

$$\phi_d = ||\mathbf{W}_d(\vec{d}^{obs} - \mathbf{G}\vec{m})||^2, \tag{7}$$

where $\mathbf{W}_d$ is a diagonal data weighting matrix, which commonly normalizes each datum by its standard deviation. In our case, the errors are unknown a priori, so we set $\mathbf{W}_d$ to be the identity matrix or use the estimate relative error levels in the data.

The model objective function ϕ_m defines the structure in the equivalent source layer and is expressed as follows,

$$\phi_m = ||\mathbf{W}_m\vec{m}||^2, \tag{8}$$

which is a discretized version of the following,

$$\phi_m = \alpha_s\int_S |m|^2 ds + \alpha_x\int_S |\frac{\partial m}{\partial x}|^2 ds + \alpha_y\int_S |\frac{\partial m}{\partial y}|^2 ds. \tag{9}$$

By using this model objective function, we require that the constructed equivalent source to be simple and smooth, which is what we need to identify uncorrelated static shifts.

The minimization of ϕ in equation 6 yields the linear system of equations,

$$(\mathbf{G}^T\mathbf{W}_d^T\mathbf{W}_d\mathbf{G} + \beta\mathbf{W}_m^T\mathbf{W}_m)\vec{m} = \mathbf{G}^T\mathbf{W}_d^T\mathbf{W}_d\vec{d}^{obs}. \tag{10}$$

The solution of equation 10 with a given regularization parameter β yields the equivalent source.

To remove static shift, we carry out a search for the optimal β that is consistent with the noise in the data set. Based on the linear inverse theory, finding the optimal β is equivalent to quantifying the noise level in the data. The predicted data from the equivalent source corresponding to the optimal β are therefore treated as the denoised data with static shift removed. Our numerical experiments have shown that the generalized cross-validation (GCV) (Wahba, 1990) works well for finding the regularization parameter.

Transformation from impedance responses to electric field

In CSEM (MacLennan and Li, 2013), one can use the electric field directly in equivalent source processing. However, only impedance data are available in MT. We must devise a way to derive electric field from impedance data. This can be accomplished by estimating a smooth background conductivity model. The magnetic field from this conductivity model is calculated by carrying out a forward modeling. Then we can extract electrical field from the impedance data. The assumption is that the magnetic field is much less affected by the near-surface conductors that give rise to the static shift.

The relationship between electric and magnetic field is given by,

$$\vec{E}_h = \mathbf{Z}\vec{H}_h, \tag{11}$$

where $\vec{E}_h$ are the horizontal components of electric field, $\vec{H}_h$ the horizontal components of magnetic field, $\mathbf{Z}$ the impedance

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tensor. Two separate forward calculations using two different external magnetic polarizations are required to generate the required magnetic field. Substituting the simulated magnetic fields into equation 11 yields the electrical fields, which can then be processed using the equivalent source technique. After processing, the denoised electrical fields are combined with the simulated magnetic field to recover the denoised impedance data.

This process can be simplified when we use a 1D layered conductivity as the background structure. For H_x polarization, we can set $H_x = 1$, $H_y = 0$ on the surface and substitute into equation 11 to yield,

$$\vec{E}_h^1 = [Z_{xx}, Z_{yx}]^T. \tag{12}$$

Therefore, two components of the impedance tensor can be treated as a scaled version of the electrical field.

For H_y polarization, we set $H_x = 0, H_y = 1$ on the surface and substitute into equation 11 to yield,

$$\vec{E}_h^2 = [Z_{xy}, Z_{yy}]^T. \tag{13}$$

In equations 12 and 13 we treat the components of the impedance tensor as if they are electric field under the assumption of a 1D background. We can then apply the proposed equivalent source technique to remove the static shift.

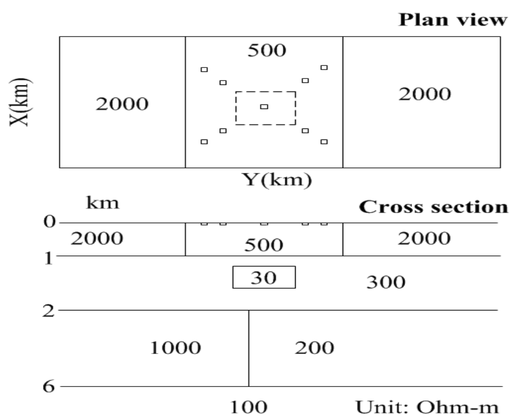

Figure 1: A synthetic 3D model simulating the presence of static shift. The nine small squares in the plan view are positions of near-surface small conductors. The dashed square in the plan section indicates the location of the deeper target.

SYNTHETIC EXAMPLE

Figure 1 shows a 3-D synthetic model designed to simulate MT data affected by static shift. The cross section shows a conductive 3D block embedded in a 2D background. The block has a resistivity of 30 $\Omega \cdot m$ and dimension of $1 \times 1 \times 0.9$ km^3. The plan section shows nine shallow conductive rectangular prisms with 10 $\Omega \cdot m$ resistivity and dimension of $100 \times 100 \times 50$ m^3 scattered around the center of this model. These conductors generate the static shift in data acquired at stations near them.

We calculate the model response using a secondary-potential formulation (Farquharson et al., 2002). Both "clean data" without static shift and "noisy data" with the effects are computed for comparison purposes. The latter are processed using the equivalent source technique.

As an illustration, we first display the results for $f = 0.05$ Hz and H_x-polarization. Figure 2 is the corresponding GCV curve. The red circle indicates the optimal value of the regularization parameter. Figure 3 shows the constructed equivalent source. Figure 4 shows the comparison of clean, noisy and denoised E_x. Compared with the clean data, noisy data are mainly distorted in the central area. After processing using the equivalent source technique, the denoised data appear an excellent representation of the clean data, demonstrating that the equivalent source technique can be effective in removing static shift in MT data.

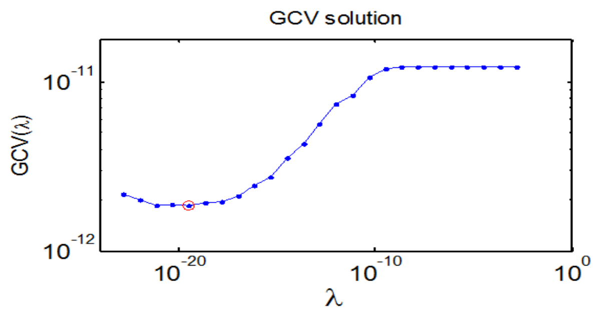

Figure 2: The GCV curve for $f = 0.05$ Hz and H_x-polarization. The red circle indicates the optimal value of λ.

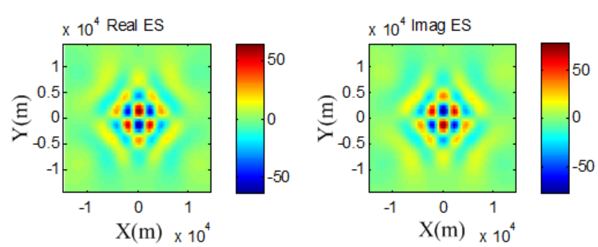

Figure 3: The constructed equivalent source for $f = 0.05$ Hz and H_x-polarization. On the left is the real part; on the right is the imaginary part.

The aforementioned process was repeated for all data from different frequencies to obtain the complete set of de-noised impedance data. Figure 5 is the comparison of apparent resistivity curves from one station. We can observe a vertical shift between clean and noisy data, which is precisely the static shift effect caused by the shallow conductors. We also observe that the denoised apparent resistivity curve has been largely shifted back toward the clean data after processing, which further demonstrates that this method is capable of dealing with the static shift consistently across different frequencies. Figure 6 presents the comparison of apparent resistivity profiles of plan view of $f = 3.2Hz$. Both the static shifts in ρ_{xy} and ρ_{yx} maps are largely removed after processing. However, this processing smoothes the signal spatially to some extent, as can be seen by comparing the clean data with the denoised data.

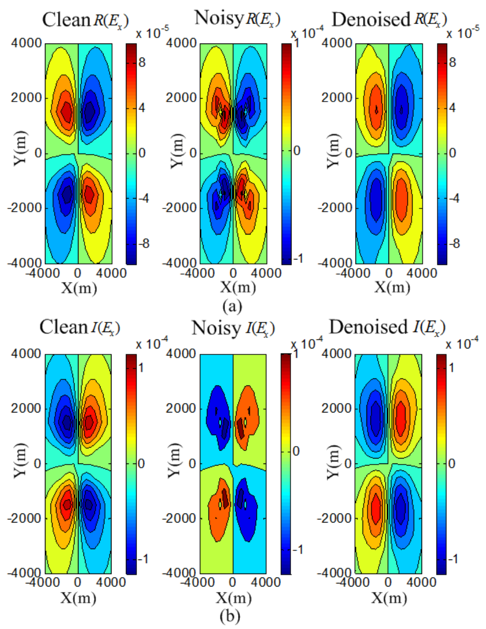

Figure 4: Comparison of clean, noisy and denoised E_x for $f = 0.05$ Hz and H_x-polarization. (a) comparison of the real parts; (b) comparison of the imaginary parts. Units are in V/m.

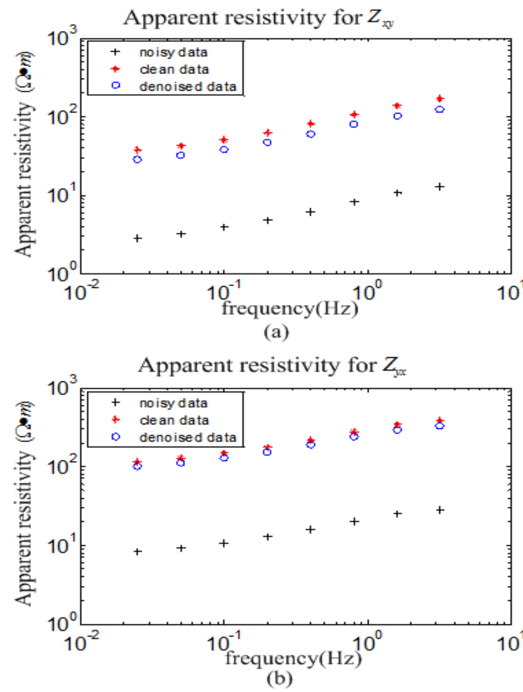

Figure 5: Comparison of apparent resistivity curves from clean, noisy and denoised data for 8 frequencies. (a) comparison of ρ_{xy} (b) comparison of ρ_{yx}.

CONCLUSIONS

We have developed a technique for removing static shift in MT data in a general 3D environment. The technique first transforms impedance data into to electric field components. It then processes the electric field by using the equivalent source technique to remove the static shift at each frequency. Finally, the denoised electric fields are converted back to obtain denoised impedance data. The technique is illustrated with a synthetic example and shown to perform well in this case. This method has the advantage that we can remove static shift starting from the MT data itself. Further testing is required to fully understand the applicability and limitations of the method in practice.

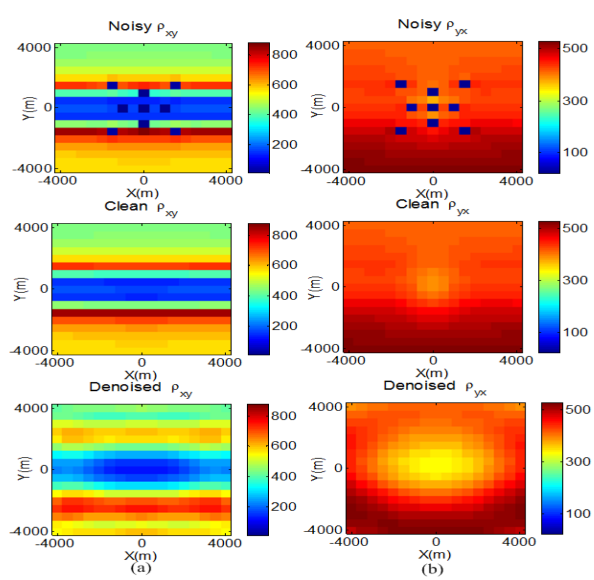

Figure 6: Comparison of apparent resistivity profiles from clean, noisy and denoised data for $f = 3.2Hz$. (a) comparison of ρ_{xy}; (b) comparison of ρ_{yx}. Units are in $\Omega \cdot m$.

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EDITED REFERENCES

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REFERENCES

Berdichevsky, M. N., and V. I. Dimitriev, 1976, Distortion of magnetic and electric fields by near-surface lateral inhomogeneities: Acta Geodaetica, Geophysica et Montanistica Academy of Science Hungary, **11**, no. 3-4, 447–483.

Bibby, H. M., T. G. Caldwell, and C. Brown, 2005, Determinable and non-determinable parameters of galvanic distortion in magnetotellurics: Geophysical Journal International, **163**, no. 3, 915–930, http://dx.doi.org/10.1111/j.1365-246X.2005.02779.x.

Bostick, F. X., 1986, Electromagnetic array profiling (EMAP): 56th Annual International Meeting, SEG, Expanded Abstracts, 60–61.

Dampney, C. N. G., 1969, The equivalent source technique: Geophysics, **34**, 39–53, http://dx.doi.org/10.1190/1.1439996.

deGroot-Hedlin, C., 1991, Removal of static shift in two dimensions by regularized inversion: Geophysics, **56**, 2102–2106, http://dx.doi.org/10.1190/1.1443022.

Farquharson, C. G., D. W. Oldenburg, E. Haber, and R. Shekhtman, 2002, An algorithm for the three-dimensional inversion of magnetotelluric data: 72nd Annual International Meeting, SEG, Expanded Abstracts, 649–652.

Groom, R. W., and R. C. Bailey, 1989, Decomposition of magnetotelluric impedance tensors in the presence of local three-dimensional galvanic distortion: Journal of Geophysical Research: Solid Earth, **94**, no. B2, 1913–1925, http://dx.doi.org/10.1029/JB094iB02p01913.

Jiracek, G. R., 1990, Near-surface and topographic distortions in electromagnetic induction: Surveys in Geophysics, **11**, no. 2-3, 163–203, http://dx.doi.org/10.1007/BF01901659.

MacLennan, K., and Y. Li, 2013, Denoising multicomponent CSEM data with equivalent source processing techniques: Geophysics, **78**, no. 3, E125–E135, http://dx.doi.org/10.1190/geo2012-0226.1.

Ogawa, Y., and T. Uchida, 1996, A two-dimensional magnetotelluric inversion assuming Gaussian static shift: Geophysical Journal International, **126**, no. 1, 69–76, http://dx.doi.org/10.1111/j.1365-246X.1996.tb05267.x.

Sharma, P. V., 1966, Rapid computation of magnetic anomalies and demagnetization effects caused by bodies of arbitrary shape: Pure and Applied Geophysics, **64**, no. 1, 89–109, http://dx.doi.org/10.1007/BF00875535.

Sternberg, B. K., J. C. Washburne, and L. Pellerin, 1988, Correction for the static shift in magnetotellurics using transient electromagnetic soundings: Geophysics, **53**, 1459–1468, http://dx.doi.org/10.1190/1.1442426.

Wahba, G., 1990, Spline models for observational data: SIAM.